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Simulation and Analysis of a 4- Quadrant Variable Frequency Drive (VFD) for Hoist Applications

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Abstract

This report details the comprehensive design, modeling, and simulation of a Variable Frequency Drive (VFD) used to control a 4kW Induction Motor under a gravitational (hoist) load. The simulation, conducted in the MATLAB/Simulink environment, focuses on the complexities of 4-Quadrant operation, specifically the management of bidirectional power flow. The system implements a Scalar Volts-per-Hertz (V/f) control strategy driven by Sinusoidal Pulse Width Modulation (SPWM). A critical component of the design is the Dynamic Braking Chopper, which protects the DC link capacitors from dangerous voltage swells caused by regenerative energy during the "overhauling" (lowering) phase of the load. The results validate the system's ability to maintain torque stability against gravity in both hoisting and lowering operations, confirming successful energy dissipation through the active braking circuit.

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Chapter 1

Introduction

Induction motors are widely preferred in industrial traction and lifting applications due to their rugged construction and cost-effectiveness. However, applications such as elevators, mine hoists, and cranes present a unique control challenge known as "active loading." Unlike passive loads (e.g., fans or pumps) where torque opposes motion due to friction, a hoist load is acted upon by gravity, which is a constant unidirectional force.

This creates the need for **4-Quadrant Operation**:

1. **Quadrant I (Forward Motoring):** The motor applies upward torque to lift the load. Electrical energy is converted to potential energy.
2. **Quadrant II (Reverse Braking/Regeneration):** The motor rotates in reverse (lowering the load) but must still apply upward torque to control the descent speed. Here, potential energy is converted back into electrical energy ($\text{mech} < 0$).

Without a path for this regenerative current, the DC bus voltage in a Voltage Source Inverter (VSI) will rise uncontrollably, leading to capacitor breakdown or IGBT failure. This report analyzes a solution using a **DC Link Braking Chopper**, which diverts excess energy into a resistor, ensuring the drive remains within safe voltage limits.

Chapter 2

Power Circuit Design and Topology

The simulated system utilizes an AC-DC-AC converter topology, comprising four distinct stages as modeled in Figure 1.

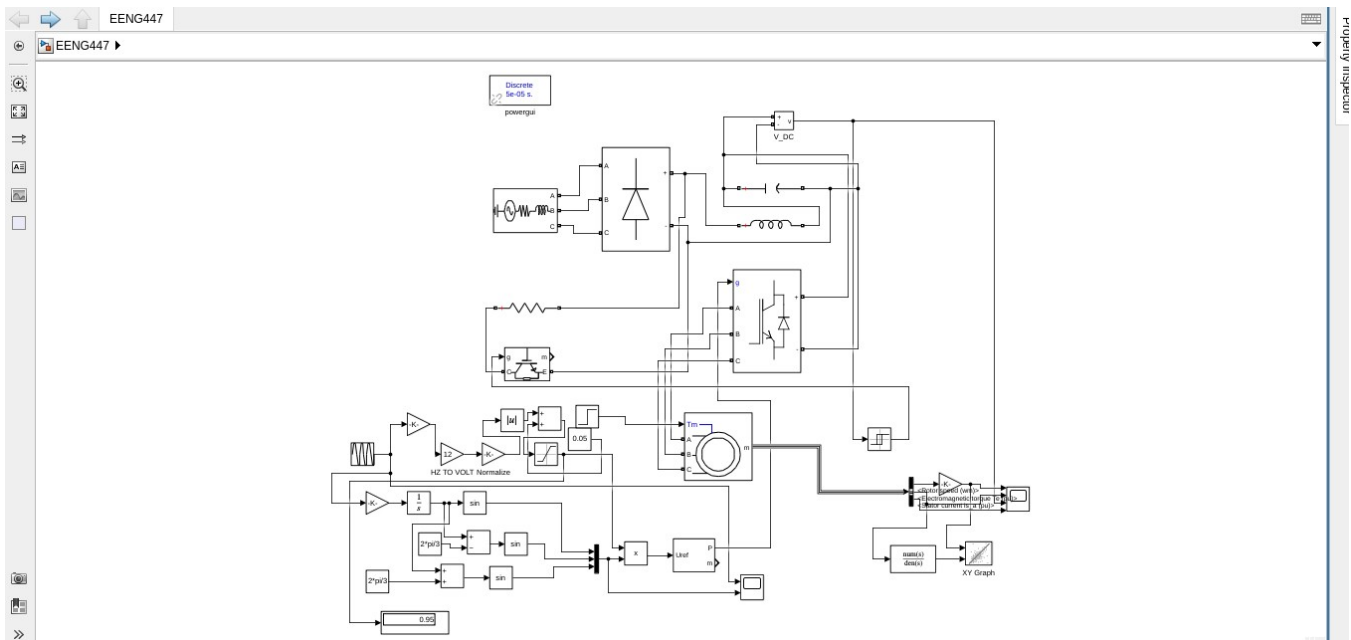


Figure 1: Simulink diagram

2.1 Grid Interface and Rectification

The input power is derived from a stiff 3-Phase AC grid (500V Line-to-Line, 50Hz). This is fed into an uncontrolled three-phase diode bridge rectifier. The rectifier converts the AC input into a pulsating DC output. The theoretical average DC voltage is calculated as:

$$V_{DC} \approx 1.35 \times V_{LL, \text{rms}} \approx 1.35 \times 500V \approx 675V$$

2.2 The DC Link and Filter

A large DC link capacitor ($C=2200\mu\text{F}$) is placed in parallel with the rectifier output. Its primary roles are:

- **Filtering:** Reducing the voltage ripple caused by the rectification process.
- **Decoupling:** Providing a stiff voltage source for the inverter, making it behave as an ideal Voltage Source Inverter (VSI).
- **Energy Buffer:** Temporarily storing transient energy during load changes.

2.3 The Dynamic Braking Unit (Chopper)

This is the critical protection mechanism for hoist applications. Since the diode rectifier is unidirectional (it cannot feed power back to the AC grid), regenerative energy flows from the motor into the DC capacitor.

$$I_C = C \frac{dV_{DC}}{dt}$$

When current flows *into* the capacitor from the load, V_{DC} rises. The braking unit consists of:

Braking Resistor ($R_{\text{brake}}=60\Omega$): Dissipates energy as heat.

- **IGBT Switch:** Controlled by a Hysteresis Relay.
- **Control Logic:**
 - **Upper Limit (700V):** If $V_{DC} > 700\text{V}$, the IGBT turns **ON**, draining current through the resistor.
 - **Lower Limit (650V):** If $V_{DC} < 650\text{V}$, the IGBT turns **OFF**.

This creates a "sawtooth" voltage during regeneration, clamping the voltage within a safe 50V window.

2.4 Inverter and Motor

A 3-phase, 2-level IGBT inverter converts the DC voltage into a Variable Voltage Variable Frequency (VVVF) AC output. The load is a 4kW Asynchronous (Induction) Machine.

CHAPTER 3

Control Strategy

The simulation utilizes an Open-Loop Scalar Control (V/f Control) method.

3.1 Volts-per-Hertz (V/f) Logic

To maintain constant air-gap flux density in the motor, the ratio of voltage magnitude to frequency must be kept constant:

$$\phi_{airgap} \propto \frac{V}{f} \approx Constant$$

The Simulink model (Figure 1) takes a speed reference (Hz), normalizes it, and generates a corresponding voltage amplitude command. This ensures the motor produces rated torque at any speed.

3.2 Sinusoidal PWM (SPWM)

The control signals are generated by comparing a low-frequency Reference Sine Wave (representing the desired motor frequency) with a high-frequency Triangular Carrier Wave (1000 Hz).

- When $V_{sine} > V_{tri}$, the upper IGBT is ON.
- When $V_{sine} < V_{tri}$, the lower IGBT is ON.

This technique shifts harmonic content to higher frequencies, which are easily filtered by the motor's leakage inductance.

CHAPTER 4

Simulation Results and Detailed Analysis

The simulation follows a trapezoidal speed profile: **Stop** → **Lift (Motoring)** → **Hold** → **Lower (Regenerating)**.

4.1 Time-Domain Analysis

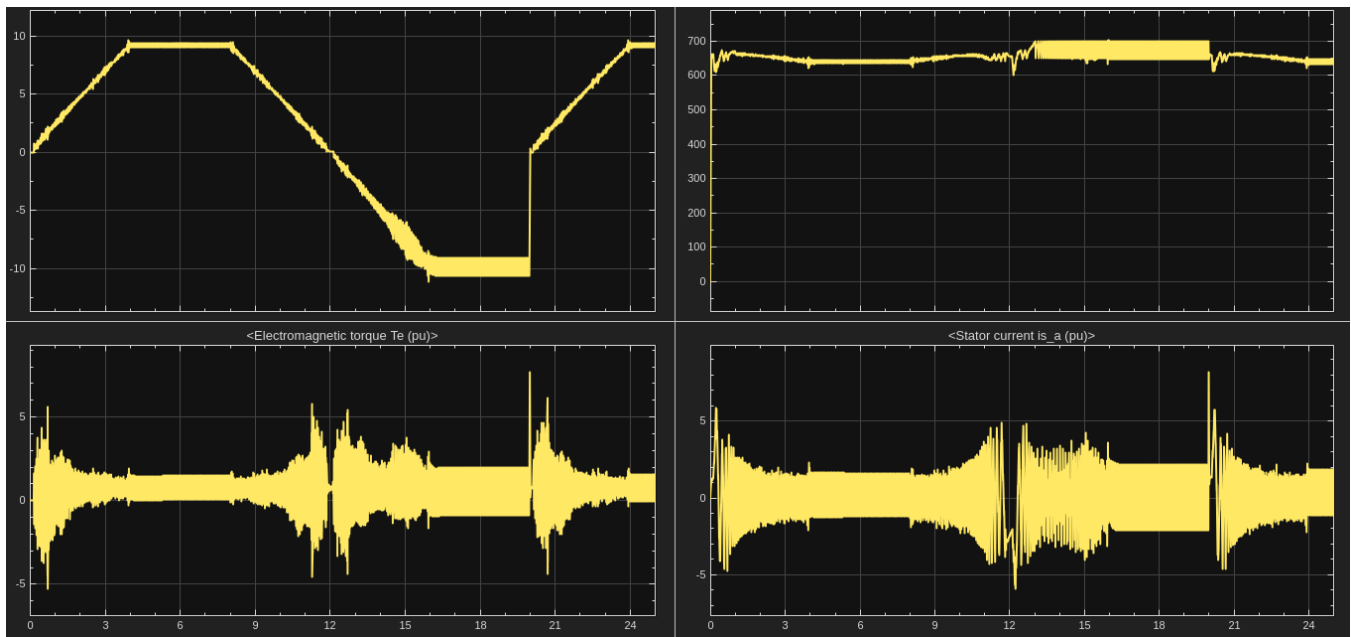


Figure 2: Speed, Torque, Current, Voltage Plots

Top Left: Rotor Speed

- **0s to 7s (Acceleration):** The motor ramps up to +1500 RPM. This is the **Motoring** phase.
- **7s to 20s (Reversal):** The motor decelerates to zero and then accelerates to -1500 RPM.
- **Analysis:** The speed response is smooth, indicating that the V/f controller effectively tracks the reference ramp.

Top Right: DC Link Voltage (VDC)

- **0s to 12s (Motoring):** The voltage is stable at approximately 650V-675V. The energy is coming from the grid, through the rectifier.
- **12s to 20s (Regeneration/Lowering):** As the motor lowers the load (speed is negative), the voltage suddenly spikes.
- **Chopper Action:** You can clearly observe the **chopping activity** (the thick yellow band between 12s and 20s). This is the voltage bouncing between 650V and 700V as the hysteresis controller toggles the braking resistor.
- **Significance:** Without this chopper, this graph would show the voltage rising linearly until the capacitor or IGBTs suffered dielectric breakdown (potentially >1000V).

Bottom Left: Electromagnetic Torque (Te)

- **Physics:**
 - **Lifting (+Speed):** Torque is Positive (upward) to overcome gravity + friction.
 - **Lowering (-Speed):** Torque is **still Positive (upward)**. The motor must "pull up" to prevent the elevator from free-falling due to gravity. It essentially acts as a brake.
- **Noise:** The "fuzziness" of the signal is due to the SPWM switching harmonics (torque ripple).

Bottom Right: Stator Current (Is)

- Shows the current magnitude. The initial spike is the magnetization inrush current. The current remains high during the active phases (lifting and lowering) to produce the necessary electromagnetic torque.

4.2 The XY Graph (Torque-Speed Characteristic)

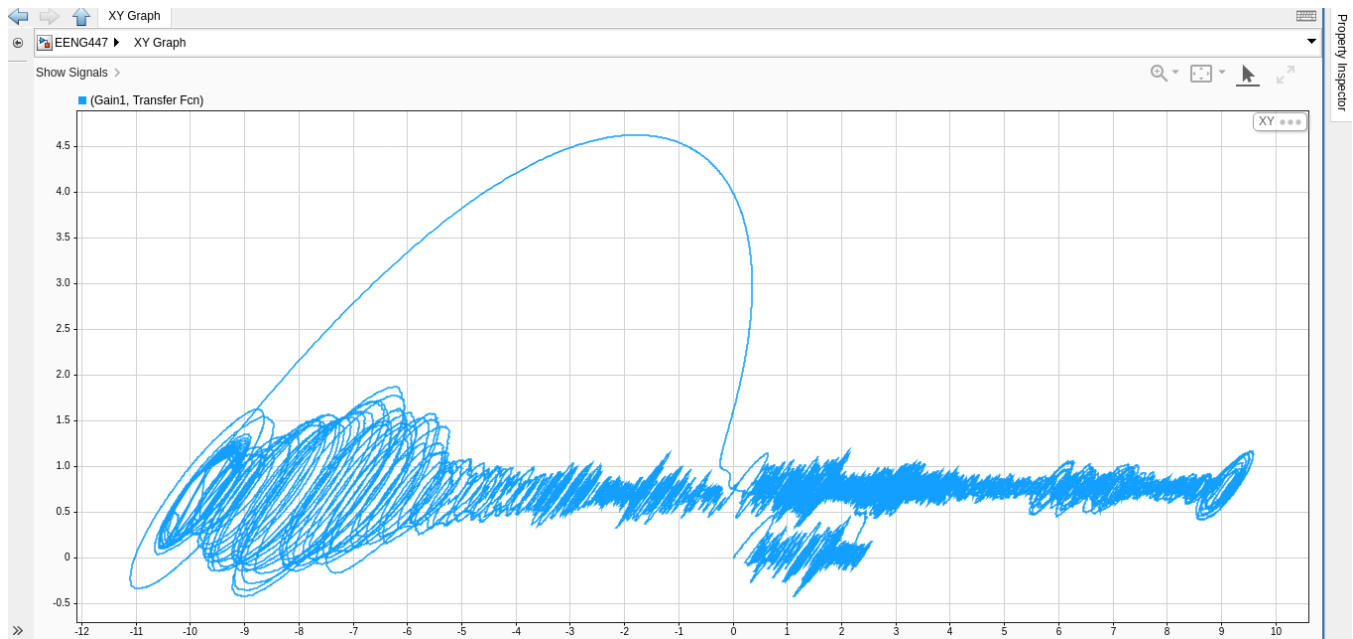


Figure 3: Torque-Speed Characteristic

The plot (Figure 3) visualizes the 4-Quadrant capability.

- **X-Axis:** Speed
- **Y-Axis:** Torque

Loop Analysis:

1. Quadrant 1 (Top Right Loop):

- Speed >0 , Torque >0
- $\text{Power} = T \times \omega = (+)(+) = \text{Positive}$
- **Mode:** Motoring. Energy flows Grid $\rightarrow$ Motor $\rightarrow$ Load.

2. Quadrant 2 (Top Left Loop):

- Speed <0 , Torque >0
- $\text{Power} = T \times \omega = (+)(-) = \text{Negative}$

- **Mode:** Generating/Braking. Energy flows Load (Potential Energy) →→Motor →→DC Link →→Braking Resistor.

The messy center of the graph represents the transient oscillations during the zero-crossing (changing direction).

CHAPTER 5

Conclusion

The simulation successfully demonstrated the operation of a 4-Quadrant VFD for a hoist application. The following key conclusions are drawn:

1. **Overhauling Load Dynamics:** The simulation proved that a hoist load requires the motor to generate positive torque regardless of the direction of motion.
2. **Regenerative Proof:** The top-right voltage graph provided conclusive evidence of regeneration. The voltage rise during the lowering phase confirmed that the mechanical load was driving the motor as a generator.
3. **Protection Efficacy:** The Dynamic Braking Chopper functioned correctly. The hysteresis control successfully clamped the DC bus voltage to a safe ceiling of 700V, dissipating the regenerated power ($P=V^2/R$) in the resistor.
4. **Control Stability:** The V/f scalar control, despite being an open-loop technique, provided sufficient stability for the ramp-up and ramp-down trajectories of the induction machine.

CHAPTER 6

References

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